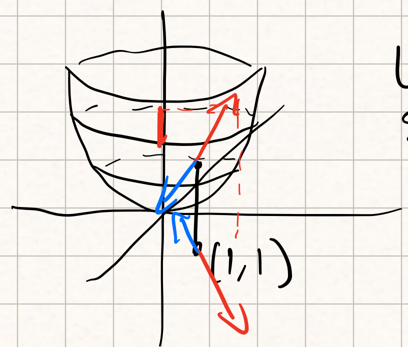
Transpose: $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$

$f: \mathbb{R}^2 \to \mathbb{R}$ $\quad f(x,y) = x^2 + y^2$ $\quad$ "Smooth" $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial y^2}, \frac{\partial^2 f}{\partial x \partial y}$ exist and are continuous



What is "the first derivative" of f? Should take vectors in $\mathbb{R}^2 \to \mathbb{R}$

"the derivative" of f at a point is a linear map $\mathbb{R}^2 \to \mathbb{R}$

"the derivative" $\in (\mathbb{R}^2)^*$

$$\underbrace{\begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \end{bmatrix}}_{\text{"the derivative"}} \begin{bmatrix} \text{"}x\text{"} \\ \text{"}y\text{"} \end{bmatrix} = \underbrace{\qquad}_{\text{rate of change of } f}$$

direction in which we're taking the derivative

"$\mathbb{R}^2$ dual takes in vector, spits out number"

$D_{(x,y)} f = \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \end{bmatrix}$ $\quad D_{(1,1)} f = \begin{bmatrix} 2 & 2 \end{bmatrix} \in (\mathbb{R}^2)^*$

$(D_{(1,1)} f) v$ is the rate of change of f at $(1,1)$ in the dir. of v

$\nabla f = \begin{bmatrix} 2x \\ 2y \end{bmatrix}$ $\quad (\nabla f)_{(1,1)} \begin{bmatrix} 2 \\ 3 \end{bmatrix} \cancel{= \begin{bmatrix} 2 \\ 2 \end{bmatrix}} \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ incoherent

$(\nabla f) \cdot (v) = \begin{bmatrix} 2 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 3 \end{bmatrix} = 4 + 6 = 10$

according to adam $Df = \left[\frac{\partial f}{\partial x} \ \frac{\partial f}{\partial y}\right]$ is the gradient

the matrix mult. $(Df)v$ is the rate of change of f in the dir. v

almost everyone else says ∇f is the gradient

Inner products (dot products)

Def: an inner product on a vector space V over $\mathbb{R}$ is a function $\underset{\text{2 vectors}}{g : V^2} \to \underset{\text{Scalar}}{\mathbb{R}}$ that satisfies the following properties

1) $g(v,v) \geq 0$
2) if $g(v,v) = 0 \quad v = 0$

} positive definite

3) $g(v,w) = g(w,v)$ } symetric

4) $g(v+w,k) = g(v,k) + g(w,k)$
5) $g(\alpha v, w) = \alpha g(v,w)$

} bilinearity

Def: the dot product is the function $\mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ given by $(x_1, \dots, x_n), (y_1, \dots, y_n) \to x_1 y_1 + \dots + x_n y_n$

Lem: the dot product is an inner product

Lem: the dot product is NOT the only inner product on $\mathbb{R}^n$

Ex: $\mathbb{R}^2$ $\left((x_1, x_2), (y_1, y_2)\right) \to x_1 y_1 + 6 x_2 y_2$

satisfies properties 1-5 of inner product